

ANDREAS PANAYI

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EDUCATION

National and Kapodistrian University of Athens (NKUA) 2024 – 2026
National Technical University of Athens (NTUA)
M.S. in Theoretical Computer Science (ALMA) GPA: 9.89/10
Selected coursework: Advanced Algorithms & Data Structures, Computational Complexity,
Graph Theory, Machine Learning, Artificial Intelligence, Algorithmic Data Science

National and Kapodistrian University of Athens (NKUA) 2020 – 2024
B.Sc. in Mathematics GPA: 8.84/10

EXPERIENCE

Algolysis Ltd Sept 2026 -
Research Engineer

- Designed and implemented robust and efficient algorithmic solutions for computationally challenging problems across distributed systems, computer graphics, computer vision and machine learning.
- Developed software and hardware solutions involving distributed storage, networking, systems programming, data-processing platforms.

Research and Innovation Foundation of Cyprus, Rinnoco Ltd Jun 2026–Jul 2026
Research and Development, Systems Engineer Intern

- Research and proof-of-concept implementation of a database compression system, working at the intersection of Big Data and Machine Learning systems development.
- Built the full system in C++20 with MLPack, PostgreSQL, and Docker, using Python for testing, and applying statistical approaches alongside supervised and unsupervised machine learning techniques.

Archimedes Research Unit, Athena Research Center Jun 2025 – Jan 2026
Intern Researcher

- Designed and implemented advanced data structures for computational geometry problems in a research setting, with applications to multi-objective optimization.
- Contributed to algorithmic work across algorithms, artificial intelligence, and data science through technical analysis and implementation.

National and Kapodistrian University of Athens 2021–2026
Teaching Assistant, Computer Science I

- Conducted undergraduate laboratory sessions for the mandatory course *Computer Science I* in the Department of Mathematics.
- Taught programming in MATLAB (2021–2024) and Python (2024–2026), as well as $\text{\LaTeX}$, computer arithmetic, and numerical linear algebra.

PROJECTS

MIT 6.5840: Distributed Systems Jul 2026 –
Go, Linux, Git

- Implementing all labs of MIT's distributed systems course in Go: MapReduce, Raft consensus, a fault-tolerant key/value store, and a sharded KV service with dynamic reconfiguration.
- Focus on concurrency, RPC, crash recovery, persistence, and testing under stress cases.

Marabu Blockchain Implementation

Completed

TypeScript, C++, Linux, SSH, GitHub

- Implemented a complete blockchain node on the Marabu protocol; deployed, tested, and debugged on a remote Vultr server via SSH and GitHub-based workflows.

Personal Webpage & Coursework Archive

Ongoing

GitHub Pages, HTML, CSS, TypeScript, XeLaTeX

- Built and maintain a portfolio site and public archive of coursework and projects; authored the introductory $\text{\LaTeX}$ and $\text{\XeLaTeX}$ notes used in university courses.

RESEARCH

Listing Even Cycles Beyond the Submodular Width Barrier

Accepted in PODS 2027

joint work with Prof. Vasilis Nakos and Dr. Hung Q. Ngo

- Developed output-sensitive algorithms for listing even-cycle queries, connecting graph algorithms with database-theoretic enumeration beyond the submodular width barrier.
- Accepted in SIGMOD/PODS 2027, the flagship conference for data management.

TECHNICAL SKILLS

Working Knowledge: Python, Go, SQL, C++

Academic Exposure: TypeScript, MATLAB, Julia, R, Haskell, Prolog

Tools: Git, GitHub, SSH, Linux, Docker, $\text{\LaTeX}$, $\text{\XeLaTeX}$

Languages: Greek (native), English (fluent)

SCHOLARSHIPS & ATHLETICS

Cyprus State Scholarships Foundation

2020 – 2026

Full scholarship for the entire duration of bachelor's and master's studies

Hellenic Bank Cyprus

2020 – 2024

Full scholarship for undergraduate studies

Athletics Distinctions

Represented Cyprus in international competitions across all age categories and earned distinctions in individual and relay events. Pancyprian champion and gold medalist in 100m, 300m hurdles, 400m, and 400m hurdles in school and club competitions.

ADDITIONAL

Military Service

Jul 2019 – Sep 2020

Served in the technical department during 14 months of military service in Cyprus